

Genomic instability and colon cancer

William M. Grady

Vanderbilt University Medical Center, 1161 21st Ave South, C2104 MCN, Nashville, TN 37232-2279

Email: william.grady@vanderbilt.edu

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Summary

Colorectal cancer affected ~135,000 people in the United States in 2001, resulting in 57,000 deaths. At the cellular level, colorectal cancer results from the progressive accumulation of genetic and epigenetic alterations that lead to the transformation of normal colonic epithelial cells to colon adenocarcinoma cells. The loss of genomic stability appears to be a key molecular and pathogenetic step that occurs early in the tumorigenesis process and serves to create a permissive environment for the occurrence of alterations in tumor suppressor genes and oncogenes. At least three forms of genomic instability have been identified in colon cancer: (1) microsatellite instability (MSI), (2) chromosome instability (i.e. aneusomy, gains and losses of chromosomal regions) (CIN), and (3) chromosomal translocations. Microsatellite instability occurs in ~15% of colon cancers and results from inactivation of the mutation mismatch repair (MMR) system by either MMR gene mutations or hypermethylation of the *MLH1* promoter. MSI promotes tumorigenesis through generating mutations in target genes that possess coding microsatellite repeats, such as *TGFBR2* and *BAX*. CIN is found in the majority of colon cancers and leads to a different pattern of gene alterations that contribute to tumor formation. CIN appears to result primarily from deregulation of the DNA replication checkpoints and mitotic-spindle checkpoints. The mechanisms that induce and influence genomic instability in cancer in general and more specifically in colon cancer are only partly understood and are consequently under intense investigation. These studies have revealed mutation of the mitotic checkpoint regulators *BUB1* and *BUBR1* and amplification of *STK15* in a subset of CIN colon cancers. The etiology of CIN in the other unexplained cases of colon cancer remains to be determined. Hopefully, discovery of the cause and specific role of genomic instability in colon cancer will yield more effective chemotherapy strategies that take advantage of this unique characteristic of cancer cells.

I. Introduction

Colorectal cancer results from the progressive accumulation of genetic and epigenetic alterations that lead to the transformation of normal colonic epithelium to colon adenocarcinoma. From the analysis of the molecular genesis of colon cancer, four central tenets concerning the pathogenesis of cancer have been established. The first is that cancer emerges via a multistep progression at both the molecular and the morphologic levels [1]. The second tenet is that the genetic and epigenetic alterations that underlie colon cancer formation

promote the cancer formation process because they provide a clonal growth advantage to the cells that acquire them. The third is that loss of genomic stability is a key molecular step in cancer formation [2]. The fourth is that hereditary cancer syndromes frequently correspond to germ line forms of key genetic defects whose somatic occurrences drive the emergence of sporadic colon cancers [3]. This fourth theme is particularly relevant regarding the role of genomic instability in colon cancer formation because of the colon cancer family syndrome hereditary nonpolyposis colon cancer (HNPCC), which results from germline

mutations in genes that encode for members of the mutation mismatch repair (MMR) system. The predisposition to cancer observed not only in HNPCC but also in other hereditary cancer syndromes caused by germline mutations in genes that regulate DNA fidelity [e.g. Li-Fraumeni syndrome (*TP53*, *CHK2*), Nijmegen breakage syndrome (*NBS1*), Bloom syndrome (*BLM*), and Ataxia telangiectasia (*ATR/ATM*)], demonstrates that inactivation of mechanisms that regulate genomic stability is a primary event in carcinogenesis.

Approximately 30% of the genes of the human genome encode for proteins that regulate DNA fidelity. Not only does this imply that the maintenance of the integrity of genomic DNA is important in eukaryotic cells but also that there are likely a variety of different mechanisms that can lead to loss of genomic DNA stability. In the majority of cancers, the mechanism responsible for aneuploidy is not known. This has led some investigators to even propose that genomic instability can arise in the absence of genetic mutations and is self-propagating in cancer [4]. However, in at least a subset of cancers, a specific genetic etiology has been found which argues that genetic alterations do mediate the loss of genomic stability in colon cancer. In fact, in 15% of colon cancers, inactivation of the mutation mismatch repair system either through aberrant CpG island methylation of *MLH1* or point mutations leads to microsatellite instability, and in some CIN colon cancers, mutations in the mitotic checkpoint regulators *BUB1* and *BUBR1* have been found [5,6]. Other potential mechanisms that may play a role in genomic instability include telomere shortening/telomerase overexpression, DNA replication checkpoint regulation, *APC* mutations, and centrosome number regulation. The specific role that genomic instability appears to play in colon cancer including the putative mechanisms and consequences will be discussed later.

Several forms of genomic instability can be observed in cancer in general. Lengauer et al. [11] recently proposed four major categories of alterations: (1) subtle sequence changes, including base substitutions, deletions, or insertions, and microsatellite instability; (2) alterations in chromosome number (aneuploidy, also termed chromosomal

instability or CIN), (3) chromosome translocations, and (4) gene amplification [2]. The genome of colon cancers, like many solid tumors, is often marked by a complex pattern of chromosome translocations that appears nearly random [7]. This form of genomic instability, 'translocation instability,' is likely mediated by the same mechanisms that cause aneuploidy in at least some cancers, but its role in colon tumorigenesis is still being defined [2]. In a recent analysis of 17 colon cancer cell lines with comparative genomic hybridization (CGH) and spectral karyotyping (SKY), 12% of the lines ($N = 2/17$) showed genomic instability that was predominantly characterized by chromosomal translocations [8]. Finally, gene amplification occurs infrequently in colon cancer but does appear to play a role in driving stage transitions in at least some colon cancers [9].

Virtually all colon cancers display either microsatellite instability (MSI) or chromosome instability (CIN) suggesting that genomic instability is not only common but also central to colon cancer formation. In fact, there is an inverse relationship between CIN and MSI; MSI cancers are diploid and show normal rates of gross chromosomal aberrations whereas CIN cancers, which are mutation mismatch repair (MMR) proficient, are aneuploid and show increased rates of chromosomal losses and gains [10,11]. The fact that CIN and MSI are usually exclusive argues against genomic instability being a nonspecific event in tumor formation and rather suggests genomic instability drives the tumorigenic process probably by permitting the accumulation of mutations that result in a cancer. In support of this concept of a permissive effect of genomic instability on the promotion of tumorigenesis, human epithelial cells have been transformed *in vitro* through the serial introduction of carefully selected oncogenes without inducing widespread genomic instability [12]. Thus, it appears that the genomic instability that is nearly universal in colon cancers may indicate that in naturally occurring tumors, the process of mutagenesis is random and stochastic and is only successful in generating cancers when it is coupled with the darwinian evolution of these mutant clones in the context of genomic DNA hypermutability.

II. Genomic instability in the adenoma-carcinoma sequence

The evolution of normal epithelial cells to adenocarcinoma usually follows a predictable progression of histologic changes and concurrent genetic and epigenetic changes. These alterations provide a growth advantage and lead to clonal expansion of the altered cells. In a manner reminiscent of darwinian evolution, subsequent alterations with waves of clonal expansion then occur as a consequence of progressive events that provide other growth advantages to the cells such as loss of cell contact inhibition. Predictably, the progressive accumulation of genetic alterations can be identi-

fied in more histologically advanced colon neoplasms [13] (see Figure 1a).

The earliest identifiable lesion in colon cancer formation is the aberrant crypt focus (ACF). The true neoplastic potential of this lesion is still unclear, but it does appear that some of these lesions harbor mutations in *K-RAS* or *APC* and can progress to frank adenocarcinoma. In particular, dysplastic aberrant crypt foci frequently carry mutations in *APC* and appear to have the highest potential for progressing to colon cancer. The increased likelihood of ACF's with mutant *APC* progressing to cancer may not only result from over-activation of the Wntless/Wnt signaling pathway but also from the induction of

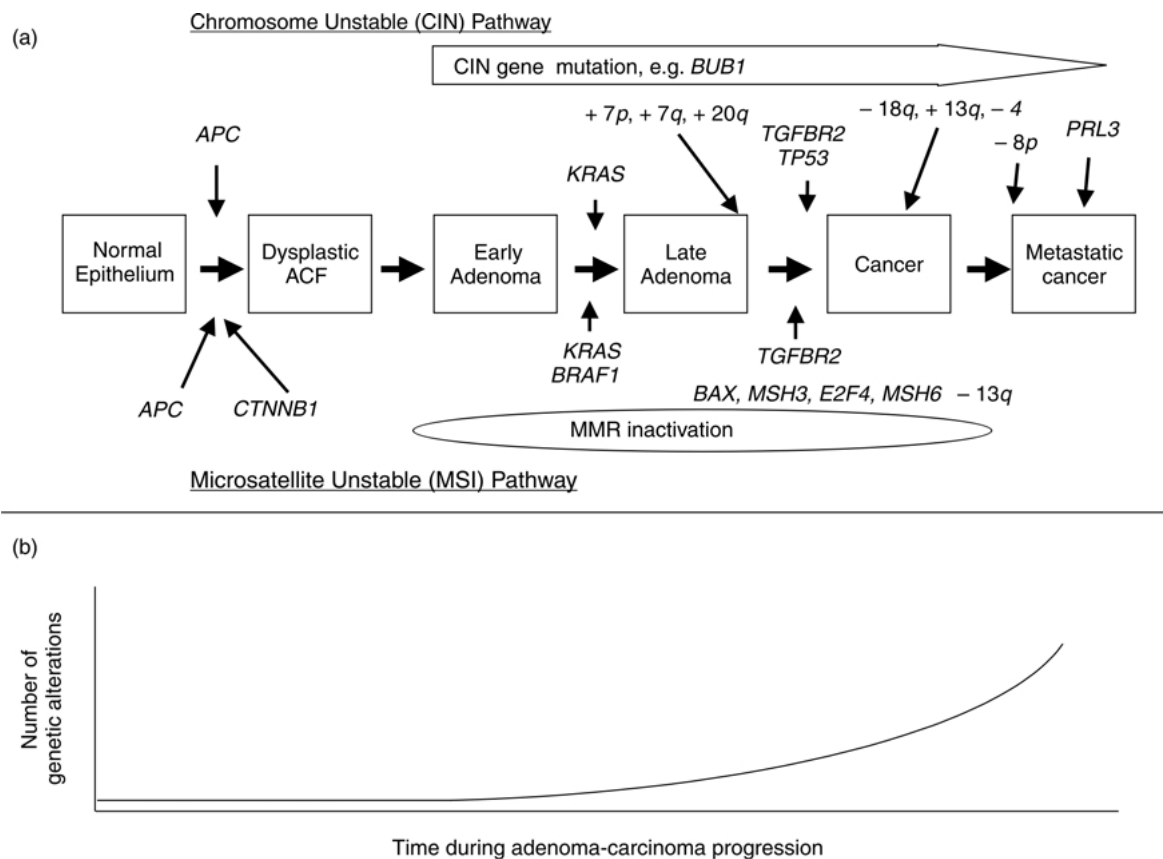


Figure 1. (a) Specific genetic alterations arise at defined stages during the adenoma-carcinoma sequence and the pattern of alterations and genes affected are often unique to tumors arising via the chromosome unstable pathway (CIN) or the microsatellite unstable (MSI) pathway. Gains and losses of chromosome arms and specific genetic alterations are shown.

(b) An increased frequency of genetic alterations compared to normal colon mucosa can be demonstrated in early colon adenomas, and this frequency increases further through the stages of more advanced colon cancer development. The upsloping line in this graph represents this increasing frequency.

chromosome instability. APC has multiple protein binding domains including an EB1 binding domain in its C-terminal end. EB1 associates with the growing ends of cytoplasmic and spindle microtubules as well as centrosomes, and the yeast homologue of EB1 is required for a microtubule-dependent cytokinesis checkpoint [14]. Fodde et al. [15] demonstrated in mouse ES cells homozygous for the mutant *Apc*^{MIN} or *Apc*1638T alleles chromosomal instability with whole-genome increment changes resulting in polyploidy. This ‘chromosomal instability’ is a consequence of abnormal mitotic spindles that form in these cells because of an abundance of microtubules that inefficiently connect with kinetochores [15,16]. However, although these provocative findings suggest that truncating mutations in *APC* induce the genomic instability observed in colon adenomas and colon cancers, the significance of these observations is still unclear. The chromosomal instability seen in *Apc* mutant cells is polyploidy rather than gains or losses of individual chromosomes, which is what is seen in colon cancer [17]. Furthermore, some well-characterized colon cancer cell lines with *APC* mutations are diploid and show relatively little chromosomal instability [11]. A unifying model for these discrepancies is that APC inactivation may create a permissive state that allows the evolving tumor clone to tolerate the development of true aneuploidy [17].

The demonstration that mutations in *APC*, the gatekeeper of the adenoma-carcinoma sequence, can cause CIN supports genomic instability playing a role in colon cancer formation even from its initiation. Further support for an active role of genomic instability in colon cancer initiation comes from the observation that both CIN and MSI occur frequently in adenomas [18–23]. Stoler et al. [24] have demonstrated the presence of approximately 11,000 genomic alterations in colon adenomas using inter-(simple sequence repeat) PCR. Furthermore, Ried et al. [22] found a stepwise increase in the average number of copy alterations using comparative genomic hybridization (CGH) as adenomas progressed from low-grade to high-grade and then finally to carcinomas [22,25,26] (see Table 1). Thus, the timing of the loss of genomic stability, either chromosomal instability (CIN) or microsatellite instability (MSI), appears to be after adenoma initiation

but before progression to frank malignancy [19–23]. In addition to the findings of Ried et al. [22], Shih et al. [27] demonstrated allelic imbalance of chromosomes 1p, 8p, 15q, and 18q in 10, 19, 28, and 28% of small adenomas (1–3 mm in size) respectively, providing more evidence of chromosomal instability in the earliest detectable adenomas. Based on the observation of LOH events in adenomas and using mathematical modeling methods, Nowack et al. [28] postulate that genomic instability occurs very early in the colon tumorigenesis process even preceding *APC* mutations (see Figure 1b).

III. Genomic instability in colon cancer

III.A. Microsatellite instability (MSI)

III.A.1. Overview of MSI and colon cancer. The DNA mismatch repair system (also known as the mutation mismatch repair (MMR) system) consists of a complex of proteins that recognize and repair basepair mismatches that occur during DNA replication. MSI occurs as the consequence of inactivation of the mutation mismatch repair system and is recognized by frameshift mutations in microsatellite repeats located throughout the genome. Inactivation of the MMR system due to germ line gene defects accounts for the colon cancer family syndrome, hereditary nonpolyposis colon cancer syndrome (HNPCC). Somatic inactivation of the mismatch repair system additionally gives rise to approximately 15% of sporadic colon cancers. In either instance the resulting colon cancers display the phenotype of microsatellite instability. The demonstration of microsatellite unstable cancers is generally performed by assaying for alterations at microsatellite loci that are particularly frequently mutated in the setting of MMR inactivation. Since many colon cancers demonstrate frameshift mutations at a small percentage of microsatellite repeats, the designation of a colon tumor as showing microsatellite instability depends on the detection of at least two unstable loci out of five from a panel of loci that were selected by a National Cancer Institute consensus conference [29].

The MSI that results from loss of MMR activity effects mono-, di-, and tri-nucleotide tracts pre-

Table 1. CIN in colon cancer formation

CR [#]	Low-grade adenoma (%)	Adenoma (any grade) (%)	Cancer (%)	MSI (%) [*]	MSS (CIN) (%) [*]	Primary (%) [*]	Metastases (%) [*]
1p		18↓	20↓				
7p	7↑	33↑	34–50↑	0	33↑	10↑	10↑
7q	0	25↑	31–34↑	0	33↑	30↑	30↑
8p	10↓	7–12↓	25–44↓	0	0	10↑ & 30↓	10↑ & 30↓
8q	0–15↑	16↑	21–53↑				
12q	0	8↑	6↑	0	0	20↑	0
13q	0–15↑	8–34↑	25–56↑	16.7↓	41.7↑	30 & 10↓	50↑
15q	5↓	12↓	31↓				
17p	0–30↓	11–29↓	36–56↓				
18q	0–10↓	9–19↓	37–58↓	0	25↓	50↓	90↓
20q	0–10↑	8–33↑	19–75↑	0	25↑	50↑	40↑
Involved Arms	7/47	21/47	32/47	3/47	22/47	34/47	35/47

Patterns of frequent loss (↓) and gain (↑) in colon cancers during different stages of tumor progression and in MSI vs. CIN cancers shown as percentage of tumors with involvement for selected chromosomes. The data in this table is derived from five different studies (Rooney et al. [23]; Ried [74]; Hermesen [25]; Hoglund [121].

[#]CR = chromosomal region, MSI = microsatellite instability, MSS = microsatellite stable, CIN = chromosome instability

^{*}Data derived from Rooney et al. [23].

Adapted from Grady and Markowitz [13].

dominantly. Cell lines from these tumors also show up to a 1,000 fold increased mutation rate at expressed gene sequences, and in particular show instability of short sequence repeats with expressed sequences [30]. Genes that possess such 'microsatellite-like' repeats in their coding regions appear to be the targets relevant to carcinogenesis. Indeed, frequently, many genes that possess microsatellite repeats are observed to be mutated in MSI colon cancers. The relationship between the microsatellite mutator pathway and other genetic alterations frequently found in colon cancer is only partially understood. Alteration of the Wnt/Wingless pathway can be observed in tumors irrespective of MSI [31]. Mutations in *APC* and *CTNNB1* can be found in 21% and 43% of MSI tumors, respectively [32,33]. In addition, the incidence of *K-RAS* mutations appears to be as high as 22–31%, which is similar to the incidence observed in microsatellite stable (MSS) colon cancers [34,35]. Mutations in *TP53* do appear to be less frequent in MSI cancers than in MSS cancers. The mutation incidence in MSI colon cancers has been demonstrated to range between 0–40%, whereas the incidence in MSS tumors is between 31–67% [10,32,34]. Of interest, monoallelic and biallelic *BAX* mutations are found fre-

quently in MSI colon cancers and may serve to replace the role of mutant *TP53* in colon carcinogenesis. Thus, the microsatellite mutator pathway appears to be initiated through changes in the Wnt/Wingless pathway and share some alterations with the MSS colon cancer pathway. However, other events, such as *TP53* and *TGFBR2* mutations, occur at different frequencies in the MSI versus the MSS pathway.

The MSI tumor formation process has been termed the microsatellite mutator phenotype and is a pathway to tumor formation that is distinct from that seen in colon cancers that are microsatellite stable [36–38]. The most frequently targeted gene for mutation in this pathway is the TGF- β receptor type II tumor suppressor gene (*TGFBR2*). Other less frequently targeted genes include the activin type II receptor (*ACVR2*); *BAX*; *SEC63*, a human homologue of a yeast DnaJ-like endoplasmic reticulum translocon component protein gene; *AIM2*, an interferon-inducible gene; *MSH3* and *MSH6*, DNA mismatch repair proteins; *RIZ*, the retinoblastoma protein-interacting zinc finger gene; and *CDX2*, an intestinal homeobox factor [37,39–42]. *CTNNB1* mutations are also found in 25% of MSI colon cancers but are not found in MSS cancers. Of note,

these mutations are transversions, rather than frameshift mutations [43]. Importantly, MSI and the subsequent target gene mutations appear to occur throughout the adenoma-to-carcinoma progression. The timing of many of these events during tumor formation remains to be mapped, but preliminary studies have shown they occur at distinct phases of tumor progression. For example, *TGFBR2* mutations have been mapped to the acquisition of the malignant phenotype suggesting that loss of wild-type *TGFBR2* is important for the adenoma to carcinoma transition [19] (see Figure 1a). Thus, MSI creates a favorable state for accumulating mutations in vulnerable genes that control critical biological activities of the cell, and these alterations can be predicted to ultimately lead to the generation of colon cancers. Although these mutations are likely occurring constantly after MSI has been established, the observation that the mutations are detected at particular stages of tumor development and also have tumor specificity is consistent with the concept that the genomic instability alone is not sufficient for the development of these tumors but rather leads to genetic alterations that are selected for during waves of clonal expansion [19,44,45].

An appreciation of the clinical significance of MSI in colon cancer has predictably lagged behind an understanding of the molecular basis of MSI. The majority of the studies done to date have attempted to assess the impact of MSI on clinical outcome. Several retrospective studies have been performed which have shown mixed results regarding the effect of MSI on prognosis. However, the most recent and largest of these studies suggest MSI and/or mutant *TGFBR2* is associated with an improved prognosis for stage II and stage III colon cancer patients [46–50]. Nonetheless, until an adequately powered, prospective trial is performed, it is premature to arrive at any conclusions regarding the clinical behavior of MSI tumors.

III.A.2. Mechanisms for microsatellite instability.

Alterations in at least six of the genes that encode proteins involved in the MMR system have been identified in either HNPCC or sporadic colon cancer. These genes include *MSH2*, *MSH3*, *MSH6*, *MLH1*, *PMS1*, and *PMS2*. Study of the

biochemistry of the MMR proteins has revealed that recognition of the base-base mismatches and insertion/deletion loops is performed by a heterodimer of either MSH2 and MSH6 or MSH2 and MSH3. Of interest, the MSH2-MSH3 heterodimer preferentially recognizes insertion/deletion loops and thus cannot compensate for loss of MSH6. Consequently, cancers arising with a loss of MSH6 function display microsatellite instability only in mononucleotide repeats [51]. The MLH1, PMS2, and PMS1 proteins appear to operate primarily in performing the repair of the base-base mismatches and insertion/deletion loops. A heterodimer of MLH1-PMS2 operates as a ‘molecular matchmaker’ and is involved in executing the repair of the mismatches in conjunction with DNA-polymerase δ and the replication factors proliferating cell nuclear antigen (PCNA), RPA, and RFC as well as the 5'→3' exo/endonucleases EXO1 and FEN1 and other unidentified 3'→5' exonucleases and helicases [51,52].

HNPCC related colon cancers account for 3–6% of all colon cancers, and germline mutations in *MSH2* and *MLH1* have been found in 45–70% of families that meet the Amsterdam criteria for HNPCC [53–55]. Since inactivation of both alleles of *hMSH2* or *hMLH1* is required to generate MSI, the cancers that arise in HNPCC kindreds frequently show loss of heterozygosity at the loci of these genes [56], or alternatively show somatic mutation of the sole wild-type MMR allele. The germline mutations that occur in *MSH2* and *MLH1* are widely distributed throughout either gene and are missense, deletion, or insertion mutations. These mutations result in frameshifts (60% of *MSH2* mutations and 40% of *hMLH1* mutations), premature truncations (23% of *MSH2* mutations), or missense mutations (31% of *hMLH1* mutations) [57]. The lack of a mutation hotspot has hampered the development of an inexpensive clinical assay to detect germline mutations in the genes known to cause HNPCC. Furthermore, because one wild-type allele is sufficient to maintain MMR activity, functional assays to detect MMR gene mutation carriers have not been developed for clinical use to date. However, proof-of-principle studies have demonstrated that it may be possible to develop such an assay by forcing a cell to a haploid state in which

case a mutant MMR allele could be detected [58,59]. Studies of the 15% of sporadic colon cancers that display MSI demonstrated these arose due to somatic inactivation of MMR genes, and not due to germline MMR gene mutations with low penetrance. While occasional somatic mutations of *MSH2* and *MLH1* were detected [60], the predominant mechanism for inactivating MMR unexpectedly proved to be the epigenetic silencing of the *MLH1* promoter due to aberrant promoter methylation [6,61]. The role of epigenetic alterations in genomic instability will be discussed in more detail below.

It is also likely that some of the cases of MSI, MMR wild-type tumors will be explained by inactivating mutations in as-yet-unidentified members of the MMR DNA repair system. Consistent with this prediction, Bellacosa et al. [62] identified a novel protein, MED1, that is an endonuclease that interacts with *MLH1*. MED1 is a candidate homologue of MthH, a bacterial protein that identifies and cleaves the newly replicated DNA strand by virtue of the strand's lack of adenine methylation at the palindromic d(GATC) sites. Mutations in genes such as *MED1* may explain low level MSI that has been observed in a subset of cancers. These low-level MSI cancers appear to behave more like MSS cancers but have some features consistent with MSI tumors [63].

III.A.3. Epigenetic alterations and microsatellite instability. The recent finding of aberrant *MLH1* promoter methylation in sporadic MSI colon cancers dramatically illustrated the role of epigenetic changes as potential pathogenetic alterations in cancer that can lead to genomic instability in colon cancer [6,61,64]. Aberrant methylation of the cancer genome, and associated silencing of the genes whose promoters demonstrated such methylation, has been well described at multiple genetic loci [65,66]. Reversion of the methylation using demethylating agents such as 5-deoxy-azacytidine frequently restores expression of these genes, demonstrating methylation in fact induces gene silencing. As inactivation of *MLH1* presumptively plays an initiating role in the pathogenesis of MSI colon cancers, the finding of aberrant methylation of *MLH1* in sporadic MSI colon cancers, and the

restoration of *MLH1* expression by demethylating the *MLH1* promoter in cell lines derived from such cancers, strongly suggested that such aberrant methylation could be a cause rather than a consequence of colon carcinogenesis [6,61,64]. Fine structure analysis of the methylation status of specific CpG's in the *MLH1* promoter has shown that the methylation status of small clusters of CpG's in the 5' region of the *MLH1* promoter appears to dictate the transcriptional status of the gene through interfering with CBF transcription factor binding to a CCAAT box [67,68]. Additional genetic support for a primary role for aberrant methylation in gastrointestinal carcinogenesis was provided by the finding that loss of expression of E-cadherin in association with CpG methylation of the wild-type *CDH1* allele in tumors occurring in the setting of the cancer family syndrome Hereditary Diffuse Gastric Cancer [69]. Studies of the effect of aberrant methylation of *MLH1* and *MGMT* also reveal that these epigenetic changes cooperate and even promote the genetic changes that occur during cancer formation [65,70]. Moreover, it is likely that the aberrant hypermethylation of 5' CpG dinucleotides that has been demonstrated to silence a variety of tumor suppressor genes including *CDH1*, *CDKN2A/p16*, thrombospondin-1 (*TSP1*), *MLH1*, and *GSTP1* may be similarly pathogenetic in the tumors in which these genetic alterations in these genes have been identified [6,64,65,71–73]. In particular, although mutation of *CDKN2A* has not been described in colon cancer, methylation of *CDKN2A/p16* is detected in 40% of colon cancers and has been found in not only colon cancer but also in colon adenomas [72]. This observation demonstrates that aberrant promoter methylation is occurring early in the adenoma sequence, although it does not confirm that the aberrant *CDKN2A/p16* methylation is a primary rather than a secondary event in the tumorigenesis process. Also worthy of note is recent progress that has elucidated how CpG island methylation can mediate transcriptional silencing by recruiting methyl binding proteins, MeCP2, MBD2, and MBD3, that recognize methylated sequence and recruit histone deacetylases (HDAC's). The HDAC's then induce changes in chromatin structure that impede the access of transcription factors to the promoter [65].

III.B. Chromosomal instability (CIN)

III.B.1. Overview of CIN and colon cancer.

Chromosomal instability is the most common type of genomic instability observed in colon cancer and occurs in approximately 85% of colon tumors. However, despite the high frequency of CIN in colon cancer and the fact that aneuploidy has long been appreciated as a hallmark of cancer, our understanding of the basis for this state of chromosomal disarray is still rudimentary. It has been unclear whether aneuploidy is simply a state that occurs nonspecifically during tumor formation, which the tumor cells can tolerate because of independence from the normal mechanisms that control cell growth and death, or whether it reflects an active process of chromosomal instability that is itself an important element in a tumor's development. The recognition that colon cancers display recurrent and tumor-specific chromosome abnormalities implies that this process is not random or simply accessory. Furthermore, the identification of specific patterns of chromosome gains and losses that occur during the colon adenoma to carcinoma sequence and the demonstration that CIN is an early event in tumor formation that increases with tumor progression is consistent with the idea that CIN is pathogenetic in colon cancer [22,23,25,74] (see Figure 1b and Table 1). Importantly for our understanding of CIN, Lengauer et al. [11] have demonstrated that aneuploidy is indeed the result of an abnormally high rate of chromosomal instability that persists throughout the life of the tumor. Using fluorescence *in situ* hybridization (FISH) of interphase cells with a panel of centromeric probes on colon cancer cell lines, Lengauer revealed a 10–100 fold increase in the rate of chromosome loss or gain per generation of the aneuploid cell lines compared to diploid colon cancer cell lines. These results were independently demonstrated in a study that assayed for LOH in markers around a selectable gene, *APRT*, and which showed a ten-fold increase in MMR-proficient (aneuploid) colon cancer cell lines compared to MMR-deficient (diploid) cell lines [75]. Using cell fusion and chromosome transfer techniques, Lengauer also proved that the aneuploid state does not cause CIN and that

CIN is an autosomal dominant trait, suggesting it arose from a gain-of-function mutation. Despite this insight into CIN, the mechanism(s) responsible for CIN has only been identified in a small subset of colon cancers. *TP53* mutations were initially suspected to be the leading cause of genomic instability in colon cancer, but no simple correlation between *TP53* mutation and the CIN phenotype has been shown by Lengauer or by others [10,11].

III.B.2. Mechanisms for chromosomal instability (CIN).

In an analogous approach to that used to determine the basis for MSI through the study of unicellular organisms that are MMR-deficient, the mechanisms responsible for CIN are being first unraveled through the study of yeast that show chromosomal instability. These studies have proven to be significantly more challenging than those studies that elucidated the etiology of MMR inactivation because there are many more candidate genes that can cause CIN. The list of candidate genes numbers more than 100 and includes genes employed in kinetochore structure and function; centrosome and microtubule formation and behavior; chromosome condensation; sister chromatid cohesion; and cell cycle checkpoint control [2,17]. Aneuploid cancers tend to show cytological abnormalities during mitosis including abnormal centrosome numbers, multipolar spindles, and lagging chromosomes implicating deregulation of the processes that mediate the mitotic-spindle checkpoint as a cause of CIN [17,76,77]. In addition to inactivation of proteins that regulate mitotic spindle checkpoint, inactivation of proteins that regulate DNA damage checkpoints, chromosome metabolism, and centrosome function have been directly shown in *in vitro* systems to influence chromosomal stability. Consequently, proteins involved in the regulation of the mitotic-spindle checkpoints and DNA replication checkpoints have been the most closely scrutinized as potential causes for CIN in cancer (see Figure 2).

III.B.2.a. Mitotic-spindle checkpoint genes and CIN.

Of this list of candidate mechanisms that may mediate CIN, the genes that are involved in detecting spindle-kinetochore interactions during

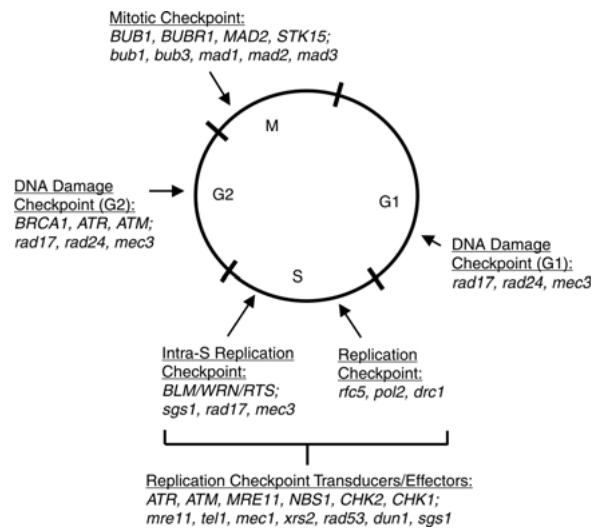


Figure 2. A schematic diagram of the cell cycle checkpoints in relation to the cell cycle phases. The DNA damage, DNA replication, and mitotic/spindle-kinetochore checkpoints are shown with a corresponding list of a subset of human and *S. cerevisiae* genes that function at each of the checkpoints. The sensor proteins and downstream signal transduction and effector proteins are listed for the DNA replication checkpoints in S phase. The proteins listed at the other checkpoints operate as sensor, transducers, or effector proteins.

mitosis are the best demonstrated to date to have a role in chromosomal instability in human cancers. Many of the proteins that regulate this checkpoint are being investigated in lower organisms, *Drosophila*, *Saccharomyces cerevisiae*, and *Xenopus*, and these studies have directly shown that the proteins encoded by the mitotic-spindle checkpoint genes control sister chromatid separation at the metaphase-anaphase transition. Importantly, the mitotic-spindle checkpoint and sister-chromatid separation pathways are well conserved allowing the results from studies in these lower organisms to be extended to studies of human cancer cells [5,78]. The *mad* and *bub* genes, in particular, are highly conserved key components that regulate the mitotic-spindle checkpoint and appear to be deregulated in cancer. The *MAD* and *BUB* gene products appear to operate as checkpoint sensors and signal transducers [5]. Activation of *MAD* and *BUB* cause cell cycle arrest through the inhibition of the APC/C, a large ubiquitin-protein ligase. The APC/C normally becomes active at the metaphase-anaphase transition resulting in the degradation of securin, a protein that inhibits anaphase. Securin interacts with separins, a class of caspase-related proteases that regulate cohesins, which create physical links

between sister chromatids. *MAD2* directly binds to the APC/C and blocks the APC/C dependent ubiquitin-mediated degradation of securin. There is also a second *MAD2*-independent mechanism that involves *BUBR1* and *BUB3* [79]. The regulation of the APC/C involves more than just *BUB* and *MAD* and integrates several inhibitory and activating inputs that are only incompletely understood at this time. For instance, studies to date indicate that securin-separin complex is regulated by an auto-feedback loop that prevents premature degradation of securin and that other proteins, such as polo kinases (PLK) mediate the activity of this complex [17]. All of these mechanisms serve to delay the progression to anaphase in order to allow enough time for the appropriate alignment of chromosomes on the metaphase plate so that the chromosomes appropriately sort during cell division.

Data showing mitotic checkpoint gene deregulation could cause CIN in colon cancer came from a landmark study by Cahill et al. in 1997. These investigators clearly demonstrated the role of mitotic spindle checkpoint genes in chromosomal instability in cancer by finding mutations in 2 members of this class of genes after recognizing aneuploid colon cancer cell lines behave in a

pattern consistent with having a spindle checkpoint defect [5]. Mutations in two genes that control the human mitotic checkpoint, *BUB1* and *BUBR1*, were found in a subset of CIN colon cancer cell lines. The mutant *BUB1* functions in a dominant negative fashion as would be predicted from cell fusion experiments with CIN cancer cell lines [11]. Mutations in other human mitotic checkpoint genes, *MAD2* and *MAD1*, have been identified in breast cancer and leukemia respectively, suggesting mutations in other genes that control chromosomal stability remain to be found in colorectal cancers [80,81]. Furthermore, *PTTG*, the 'pituitary tumor-transforming gene' and the human homologue of securin, is overexpressed in a variety of tumor types, including colon cancer [82–84]. Importantly, *MAD2*, *BUB1*, and *BUBR1* mutations generate the types of chromosome loss seen in naturally-occurring human cancers, which has not been the case in studies of other candidate CIN regulating genes such as *APC* [5,85]. Of note, despite the initial success of this candidate approach, mutation analysis of other mitotic spindle checkpoint genes, *MAD1L1*, *MAD2*, *MAD2B*, *BUB3*, *MPS1L1*, and *CDC20*, has not revealed any mutations in a panel of 19 aneuploid colorectal cancer cell lines [86].

In addition to the deregulation of proteins that directly govern the mitotic-spindle checkpoint being a candidate mechanism for CIN, abnormal centrosome number and function has been a candidate mechanism for aneuploidy since Boveri first observed centrosomes over 100 years ago [87]. Most solid cancers are characterized by centrosome amplification as well as by aneuploidy [88]. Because the centrosome plays a central role in chromosome segregation during mitosis, the centrosome amplification observed in cancers has been postulated to increase the frequency of abnormal mitoses and chromosome missegregation [89–91]. More concrete evidence of abnormal centrosome function causing chromosomal instability in colon cancer became available as the result of assessment of the genes in a locus on chromosome 20q13, which is frequently amplified in a variety of cancers including breast, colon, and pancreatic cancer [91]. Sen et al. [92] cloned *STK15/BTAK/ARK1/aurora 2*, a centrosome-associated serine-threonine kinase, from this region and detected overexpression in breast cancer cell

lines. Subsequent studies revealed *STK15* amplification and overexpression in colon cancer cell lines as well and demonstrated that over-expression of this gene can induce aneuploidy in NIH3T3 cells [91,93]. Studies in *Drosophila* and *Saccharomyces cerevisiae* indicate that *STK15* has a role in centrosome maturation, spindle assembly, and chromosome segregation and that its overexpression may lead to abnormal mitotic spindle assembly and CIN [87,94,95]. Furthermore, studies in prostate and breast cancer have demonstrated an association between aneuploidy, abnormal centrosome number, and clinically aggressive disease [89,90]. Thus, *STK15* amplification appears to be a promising potential cause of CIN. Further studies are needed to develop a better understanding of the role of centrosome amplification and centrosome-associated kinase activity in CIN in primary human colon cancer.

III.B.2.b. DNA checkpoint genes and CIN.

Genomic stability is maintained not only by checkpoints that regulate mitosis and sister chromatid separation but also by checkpoints that regulate the maintenance of DNA fidelity. In fact, cell cycle checkpoints were originally recognized as pathways that mediate cell cycle progression in response to DNA damage. Mutations that disrupt replication checkpoint genes, such as *rfc5-1*, *dpb11-1*, *mec1Δ*, *ddc2Δ*, and *dun1Δ*, substantially increase the rate of genome rearrangements [96]. These DNA damage checkpoints are traditionally considered in two classes: (1) the DNA damage checkpoints that recognize and respond to DNA damage from genotoxic agents, and (2) the DNA replication checkpoints that monitor the fidelity of replicated DNA [97]. The proteins involved in the DNA checkpoints act either as one of following: (1) sensors of DNA damage; (2) transducers of the DNA damage signal to the effector proteins; or (3) effectors that either regulate cell cycle arrest/progression, DNA repair, or both. A number of these proteins have been identified in yeast and some have already been shown to have a role in human cancer, including *TP53*, *ATM*, the ATM-related gene *ATR*, *BRCA1* and *BRCA2*, which interact with the human Rad51 homolog, and *hRAD17* [98–102] (see Table 2). In yeast, *Mec1*

Table 2. DNA replication checkpoint genes involved in human cancer. Currently identified DNA replication checkpoint genes that have been identified in humans to predispose to cancer when mutated in germline DNA. The function of the protein, gene name in yeast, homologous gene in humans, and the name of the cancer family syndrome(s) are shown

Protein function	Yeast gene	Human gene	Syndrome
Sensor: BRCT* containing	Rad9	BRCA1	Hereditary breast-ovarian cancer
Sensor: DSB* recognition/repair	Mre11	MRE11	Ataxia-telangiectasia-like disorder
Sensor: Replication proteins	Sgs1	BLM/WRN/RTS	Bloom syndrome Werner syndrome Rothmund-Thomson syndrome
Transducer: PI3-kinase (PIKK)	Mec1/Tel1	ATM	Ataxia-telangiectasia
Transducer: Effector kinase		TP53	Li-Fraumeni syndrome
Transducer: Effector kinase	Rad53	CHK2	Li-Fraumeni-like syndrome

*BRCT = Brcal carboxy terminus, DSB = double-strand break.

and Tel1, which are homologous to ATM and ATR, appear to operate as the transducer/effector components of an important DNA damage signaling network that regulates the cellular response to a variety of DNA damage events. These proteins belong to a family of protein kinases termed PIKK's (phosphatidylinositol 3-kinase-like protein kinase), and they phosphorylate several key effector proteins (e.g. Rad53p/hCHK2, Chk1p/hCHK1) in response to DNA damage. These effector proteins require additional factors to form a pentameric replication factor C-like (RFC) complex that can associate with sites of DNA damage and elicit the appropriate cellular response [103]. These DNA replication checkpoints operate during S-phase to either arrest or slow replication in response to DNA damage, and consist of both an S-phase checkpoint as well as an intra-S-phase checkpoint. These checkpoints are redundant but if both are inactivated there is a synergistic increase in the rate of genomic rearrangements. The kinds of rearrangements observed in primary human cancer are similar to those seen when these checkpoint proteins are inactivated in yeast providing indirect evidence for a role of the replication checkpoints in the genomic instability found in colon cancer [104]. Of all these checkpoint proteins, only *TP53* has been clearly shown to have a role in human colon cancer.

However, p53 has many functions and while TP53 has been demonstrated to affect colon cancer formation and to regulate the DNA damage checkpoint, it is still not clear whether *TP53*

mutation plays a role in inducing genomic instability in colon cancer. The p53 protein was initially identified as a protein forming a stable complex with the SV40 large T antigen and was originally suspected to be an oncogene [105]. Subsequent studies demonstrated that p53 is a transcription factor with tumor suppressor activity that is located at 17p13.1 and that it is mutated in 50% of primary human tumors, including tumors of the gastrointestinal tract [106]. p53 is currently appreciated to be a transcription factor that is involved in maintaining genomic stability through the control of cell cycle progression and apoptosis in response to genotoxic stress [106]. p53 normally serves to regulate cell growth and division in the context of genotoxic stress. It is expressed at very low levels in cells until it is activated by incompletely understood mechanisms by DNA damage resulting from gamma irradiation, ultraviolet irradiation, or chemotherapeutic agents [107]. Its activation results in the transcription of genes that directly regulate cell cycle progression and apoptosis. These genes include *p21^{WAF1/CIP1}*, *GADD45*, *MDM2*, *14-3-3-σ*, *BAX*, *B99*, *TSPI*, *KILLER/DR5*, *FAS/APO1*, *CYCLIN G*, and others [106]. Expression of many of these genes effectively halts DNA replication and induces DNA repair [108–111]. This function of p53 to recognize DNA damage and induce cell cycle arrest and DNA repair or apoptosis has led to p53 being called the 'guardian of the genome' [107]. However, despite substantial experimental data demonstrating p53's role in the DNA damage checkpoint, inactivation of *TP53* in the diploid

colon cancer cell line HCT116 and a human fetal fibroblast cell line, FBp53KO, failed to induce aneuploidy in these cells [112]. Thus, *TP53* inactivation alone is not sufficient to induce CIN, but may be important only in the context of other genetic alterations or in the context of genotoxic stresses not present in tissue culture conditions.

Other factors than genetic inactivation of the proteins controlling the replication checkpoints have also been proposed to play a role in inducing genomic instability in colon cancer through effects on cell cycle regulation and DNA replication. These mechanisms include TGF- β resistance, JC virus infection, and *C-MYC* overexpression [113,114]. For example, *MYC* overexpression has been implicated in attenuating the DNA damage checkpoint at G1-S phase and contributing to tumorigenesis through the induction of aneuploidy by this mechanism [115]. Nonetheless, the data supporting these mechanisms having a role in genomic instability is confined to results from *in vitro* systems and has not been generated from *in vivo* models or in primary human cancers to date.

III.B.3. Telomerase and CIN. Recently, data from animal models of cancer have demonstrated that telomere dysfunction may play a role in enabling genomic instability to occur in cancer cells. Telomeres are structures at the ends of linear chromosomes that are believed to play a fundamentally important role in regulating senescence. Conventional DNA polymerases are not able to completely synthesize chromosomal ends resulting in the gradual shortening of telomeres with successive rounds of cell divisions until a critical short length is reached that elicits the activation of cellular checkpoints similar to those activated by DNA damage [116]. In human cells, this telomere shortening culminates in activation of the Hayflick limit and the cessation of cell divisions. However, if p53 or Rb is inactivated then the cells can continue to divide and will go through a period of massive cell death termed 'cellular crisis'. Cells that survive crisis appear to activate mechanisms to maintain the telomeres, and do this most frequently by increasing the expression of telomerase, a specialized ribonucleoprotein complex that consists of a catalytic telomerase reverse transcriptase (TERT) and an RNA subunit

encoded by *TERC*. Most human cancers express telomerase, and, thus, it appears that during the evolution of normal cells to cancer cells that the cells progress through a period of severe telomere dysfunction before regaining mechanisms to maintain telomere length [117]. Now data from experiments from *Terc*^{-/-} mice suggest that this period of telomere dysfunction may cause marked chromosomal instability that promotes the formation of cancer cells [116]. Consistent with this model, human colon cancers show a peak in the anaphase bridge index, which is a measure of the number of metaphases that contain anaphase bridges, in early high-grade dysplastic lesions and less in more advanced carcinoma stages [118]. However, the identification of CIN in early colon adenomas, at a stage that precedes telomere dysfunction, suggests that telomere dysfunction cannot be the only potential mechanism that induces genomic instability in colon cancer and that its role in tumor formation may depend on the nature of concurrent genetic alterations in the tumor cells. In support of this model, *mTerc*^{-/-};Min (*Apc*^{+/-}), mice form fewer macroadenomas than mice with wild-type *mTerc*, but *mTerc*^{-/-}; *Tp53*^{-/-} mice show a high level of genomic instability similar to that seen in human cancers and reduced tumor latency in late generation mice [116,119,120]. Thus, it appears that telomere dysfunction may also contribute to genomic instability seen in cancer, although more definitive evidence of its role in human cancer remains to be shown.

IV. Conclusions

Colon cancer genetics has yielded a host of new insights and paradigms that have broadly informed the studies of most solid tumors and the role that genomic instability plays in carcinogenesis in general. Key insights that have been contributed to the field of cancer genetics include the following: (1) the identification of genomic instability early in the cancer formation process even in the premalignant phase of the adenoma-carcinoma sequence, (2) the role of DNA repair genes in genomic stability, and (3) the role of mitotic-spindle checkpoint genes in chromosomal instability. Nonetheless, many challenges remain.

A more complete understanding of the basis of chromosomal instability, aneuploidy, and aberrant methylation of the cancer genome has yet to be achieved. Furthermore, the translation of molecular genetics to new diagnostic, prognostic, and therapeutic modalities remains a challenge. The promise for the future is that this field of inquiry, which has been so fruitful to date, will continue to yield the important answers to these and other key questions.

V. Key unanswered questions

1. What are the mechanisms responsible for most cases of CIN colon cancer?
2. Is one mechanism for causing genomic instability sufficient to accomplish this, or do multiple mechanisms need to cooperate to cause genomic instability in colon cancer?
3. Is genomic instability required for primary human colon cancer initiation, progression, or both?
4. How can genomic instability be exploited from a therapeutic standpoint to develop agents that specifically target cancer cells?

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References

1. Fearon E, Vogelstein B: A genetic model for colorectal tumorigenesis. *Cell* 61: 759–767, 1990
2. Lengauer C, Kinzler K, Vogelstein B: Genetic instabilities in human cancers. *Nature* 396: 643–649, 1998
3. Kinzler K, Vogelstein B: Lessons from hereditary colorectal cancer. *Cell* 87: 159–170, 1996
4. Duesberg P, Rausch C, Rasnick D, Hehlmann R: Genetic instability of cancer cells is proportional to their degree of aneuploidy. *Proc Natl Acad Sci USA* 95: 13692–7, 1998
5. Cahill D, Lengauer C, Yu J, Riggins G, Willson J, Markowitz S, Kinzler K, Vogelstein B: Mutations of mitotic checkpoint genes in human cancers. *Nature* 392: 300–303, 1998
6. Veigl M, Kasturi L, Olechnowicz J, Ma A, Lutterbaugh J, Periyasamy S, Li G-M, Drummond J, Modrich P, Sedwick D, Markowitz S: Biallelic inactivation of *hMLH1* by epigenetic gene silencing, a novel mechanism causing human MSI cancers. *Proceedings of the National Academy of Science* 95: 8698–8702, 1998
7. Bardi G, Sukhikh T, Pandis N, Fenger C, Kronborg O, Heim S: Karyotypic characterization of colorectal adenocarcinomas. *Genes, Chromosomes, and Cancer* 12: 97–109, 1995
8. Abdel-Rahman WM, Katsura K, Rens W, Gorman PA, Sheer D, Bicknell D, Bodmer WF, Arends MJ, Wyllie AH, Edwards PA: Spectral karyotyping suggests additional subsets of colorectal cancers characterized by pattern of chromosome rearrangement. *Proc Natl Acad Sci USA* 98: 2538–43, 2001
9. Saha S, Bardelli A, Buckhaults P, Velculescu VE, Rago C, Croix BS, Romans KE, Choti MA, Lengauer C, Kinzler KW, Vogelstein B: A phosphatase associated with metastasis of colorectal cancer. *Science* 294: 1343–6, 2001
10. Eshleman J, Casey G, Kochera M, Sedwick W, Swinler S, Veigl M, Willson J, Stuart S, Markowitz S: Chromosome number and structure both are markedly stable in RER colorectal cancers and are not destabilized by mutation of p53. *Oncogene* 17: 719–725, 1998
11. Lengauer C, Kinzler K, Vogelstein B: Genetic instability in colorectal cancers. *Nature* 386: 623–627, 1997
12. Zimonjic D, Brooks MW, Popescu N, Weinberg RA, Hahn WC: Derivation of human tumor cells *in vitro* without widespread genomic instability. *Cancer Res* 61: 8838–44, 2001
13. Grady W, Markowitz S: Genomic Instability in Colorectal Cancer, *Current Opinions in Gastroenterology* 16: 62–67, 2000
14. Muhua L, Adames NR, Murphy MD, Shields CR, Cooper JA: A cytokinesis checkpoint requiring the yeast homologue of an APC- binding protein. *Nature* 393: 487–91, 1998
15. Fodde R, Kuipers J, Rosenberg C, Smits R, Kielman M, Gaspar C, van Es JH, Breukel C, Wiegant J, Giles RH, Clevers H: Mutations in the APC tumor suppressor gene cause chromosomal instability. *Nat Cell Biol* 3: 433–438, 2001
16. Kaplan KB, Burds AA, Swedlow JR, Bekir SS, Sorger PK, Nathke IS: A role for the Adenomatous Polyposis Coli protein in chromosome segregation. *Nat Cell Biol* 3: 429–432, 2001
17. Jallepalli PV, Lengauer C: Chromosome segregation and cancer: Cutting through the mystery. *Nat Rev Cancer* 1: 109–117, 2001
18. Aaltonen L, Peltomaki P, Mecklin J-P, Jarvinen H, Jass J, Green J, Lynch H, Watson P, Tallquist G, Juhola M, Sistonen P, Hamilton S, Kinzler K, Vogelstein B, de la Chapelle A: Replication errors in benign and malignant tumors from hereditary nonpolyposis colorectal cancer patients. *Cancer Research* 54: 1645–1648, 1994

19. Grady W, Rajput A, Myeroff L, Liu D, Kwon K-H, Willis J, Markowitz S: Mutation of the type II transforming growth factor- β receptor is coincident with the transformation of human colon adenomas to malignant carcinomas. *Cancer Research* 58: 3101–3104, 1998
20. Jacoby R, Marshall D, Kailas S, Schlack S, Harms B, Love R: Genetic instability associated with adenoma to carcinoma progression in hereditary nonpolyposis colon cancer. *Gastroenterology* 109: 73–82, 1995
21. Bomme L, Bardi G, Pandis N, Fenger C, Kronborg O, Heim S: Cytogenetic analysis of colorectal adenomas: Karyotypic comparisons of synchronous tumors. *Cancer Genetics and Cytogenetics* 106: 66–71, 1998
22. Ried T, Heselmeyer-Haddad K, Blegen H, Schrock E, Auer G: Genomic changes defining the genesis, progression, and malignancy potential in solid human tumors: A phenotype/genotype correlation. *Genes, chromosomes, and cancer* 25: 195–204, 1999
23. Rooney P, Murray G, Stevenson D, Haites N, Cassidy J, McLeod H: Comparative genomic hybridization and chromosomal instability in solid tumors. *British Journal of Cancer* 80: 862–873, 1999
24. Stoler DL, Chen N, Basik M, Kahlenberg MS, Rodriguez-Bigas MA, Petrelli NJ, Anderson GR: The onset and extent of genomic instability in sporadic colorectal tumor progression. *Proc Natl Acad Sci USA* 96: 15121–15126, 1999
25. Hermsen M, Postma C, Baak J, Weiss M, Rapallo A, Sciutto A, Roemen G, Arends JW, Williams R, Giaretti W, De Goeij A, Meijer G: Colorectal adenoma to carcinoma progression follows multiple pathways of chromosomal instability. *Gastroenterology* 123: 1109–1119, 2002
26. Meijer GA, Hermsen MA, Baak JP, van Diest PJ, Meuwissen SG, Belien JA, Hoovers JM, Joenje H, Snijders PJ, Walboomers JM: Progression from colorectal adenoma to carcinoma is associated with non-random chromosomal gains as detected by comparative genomic hybridisation. *J Clin Pathol* 51: 901–909, 1998
27. Shih IM, Zhou W, Goodman SN, Lengauer C, Kinzler KW, Vogelstein B: Evidence that genetic instability occurs at an early stage of colorectal tumorigenesis. *Cancer Res* 61: 818–822, 2001
28. Nowak MA, Komarova NL, Sengupta A, Jallepalli PV, Shih IM, Vogelstein B, Lengauer C: The role of chromosomal instability in tumor initiation. *Proc Natl Acad Sci USA* 21: 21, 2002
29. Boland C, Thibodeau S, Hamilton S, Sidransky D, Eshleman J, Burt R, Meltzer S, Bigas-Rodriguez M, Fodde R, Ranzani G, Srivastava S: National Cancer Institute workshop on microsatellite instability for cancer detection and familial predisposition: Development of international criteria for the determination of microsatellite instability in colorectal cancer. *Cancer Research* 58: 5248–5257, 1998
30. Eshleman J, Lang E, Bowerfind G, Parsons R, Vogelstein B, Willson J, Veigl M, Sedwick W, Markowitz S: Increased mutation rate at the *hprt* locus accompanies microsatellite instability in colon cancer. *Oncogene* 10: 33–37, 1995
31. Huang J, Papadopoulos N, McKinley A, Farrington S, Curtis L, Wyllie A, Zheng S, Willson J, Markowitz S, Morin P, Kinzler K, Vogelstein B, Dunlop M: APC mutations in colorectal tumors with mismatch repair deficiency. *Proceedings of the National Academy of Science USA* 93: 9049–9054, 1996
32. Konishi M, Kikuchi-Yanoshita R, Tanaka K, Muraoka M, Onda A, Okumura Y, Kishi N, Iwama T, Mori T, Koike M, Ushio K, Chiba M, Nomizu S, Konishi F, Utsunomiya J, Miyaka M: Molecular nature of colon tumors in hereditary nonpolyposis colon cancer, familial polyposis, and sporadic colon cancer. *Gastroenterology* 111: 307–317, 1996
33. Miyaki M, Iijima T, Kimura J, Yasuno M, Mori T, Hayashi Y, Koike M, Shitara N, Iwama T, Kuroki T: Frequent mutation of beta-catenin and APC genes in primary colorectal tumors from patients with hereditary nonpolyposis colorectal cancer. *Cancer Res* 59: 4506–4509, 1999
34. Fujiwara T, Stolker JM, Watanabe T, Rashid A, Longo P, Eshleman JR, Booker S, Lynch HT, Jass JR, Green JS, Kim H, Jen J, Vogelstein B, Hamilton SR: Accumulated clonal genetic alterations in familial and sporadic colorectal carcinomas with widespread instability in microsatellite sequences. *Am J Pathol* 153: 1063–1078, 1998
35. Olschwang S, Tiret A, Laurent-Puig P, Muleris M, Parc R, Thomas G: Restriction of ocular fundus lesions to a specific subgroup of APC mutations in adenomatous polyposis coli patients. *Cell* 75: 959–968, 1993
36. Yamamoto H, Sawai H, Perucho M: Frameshift somatic mutations in gastrointestinal cancer of the microsatellite mutator phenotype. *Cancer Research* 57: 4420–4426, 1997
37. Yamamoto H, Sawai H, Weber T, Rodriguez-Bigas M, Perucho M: Somatic frameshift mutations in DNA mismatch repair and proapoptosis genes in hereditary nonpolyposis colorectal cancer. *Cancer Research* 58: 997–1003, 1998
38. Perucho M: Cancer of the microsatellite mutator phenotype. *Biological Chemistry* 377: 675–684, 1996
39. Markowitz S, Wang J, Myeroff L, Parsons R, Sun L, Lutterbaugh J, Fan R, Zborowska E, Kinzler K, Vogelstein B, Brattain M, Willson J: Inactivation of the type II TGF- β receptor in colon cancer cells with microsatellite instability. *Science* 268: 1336–1338, 1995
40. Piao Z, Fang W, Malkhosyan S, Kim H, Horii A, Perucho M, Huang S: Frequent frameshift mutations of RIZ in sporadic gastrointestinal and endometrial carcinomas with microsatellite instability. *Cancer Res* 60: 4701–4704, 2000
41. Wicking C, Simms LA, Evans T, Walsh M, Chawengsaksophak K, Beck F, Chenevix-Trench G, Young J, Jass J, Leggett B, Wainwright B: CDX2, a human homologue of *Drosophila* caudal, is mutated in both alleles in a replication error positive colorectal cancer. *Oncogene* 17: 657–659, 1998
42. Mori Y, Yin J, Rashid A, Leggett BA, Young J, Simms L, Kuehl PM, Langenberg P, Meltzer SJ, Stine OC: Instabilotyping: Comprehensive identification of frameshift mutations caused by coding region microsatellite instability. *Cancer Res* 61: 6046–6049, 2001

43. Mirabelli-Primdahl L, Gryfe R, Kim H, Millar A, Luceri C, Dale D, Holowaty E, Bapat B, Gallinger S, Redston M: β -catenin mutations are specific for colorectal carcinomas with microsatellite instability but occur in endometrial carcinomas irrespective of mutator pathway. *Cancer Research* 59: 3346–3351, 1999
44. Parsons R, Myeroff L, Liu B, Willson J, Markowitz S, Kinzler K, Vogelstein B: Microsatellite instability and mutations of the transforming growth factor β type II receptor gene in colorectal cancer. *Cancer Research* 55: 5548–5550, 1995
45. Schwartz S, Yamamoto H, Navarro M, Maestro M, Reventos J, Perucho M: Frameshift mutations at mononucleotide repeats in *caspase-5* and other target genes in endometrial and gastrointestinal cancer of the microsatellite mutator phenotype. *Cancer Research* 59: 2995–3002, 1999
46. Gryfe R, Kim H, Hsieh ET, Aronson MD, Holowaty EJ, Bull SB, Redston M, Gallinger S: Tumor microsatellite instability and clinical outcome in young patients with colorectal cancer [see comments]. *N Engl J Med* 342: 69–77, 2000
47. Watanabe T, Wu TT, Catalano PJ, Ueki T, Satriano R, Haller DG, Benson AB, 3rd, Hamilton SR: Molecular predictors of survival after adjuvant chemotherapy for colon cancer. *N Engl J Med* 344: 1196–1206, 2001
48. Messerini L, Ciantelli M, Baglioni S, Palomba A, Zampi G, Papi L: Prognostic significance of microsatellite instability in sporadic mucinous colorectal cancers. *Human Pathology* 30: 629–634, 1999
49. Feeley K, Fullard J, Heneghan M, Smith T, Maher M, Murphy R, O'Gorman T: Microsatellite instability in sporadic colorectal carcinoma is not an indicator of prognosis. *Journal of Pathology* 188: 14–17, 1999
50. Iacopetta B, Welch J, Soong R, House A, Zhou X, Hamelin R: Mutation of the transforming growth factor-beta type II receptor gene in right-sided colorectal cancer: Relationship to clinicopathological features and genetic alterations. *Journal of Pathology* 184: 390–395, 1998
51. Jiricny J: Replication errors: Cha(lle)nging the genome. *Embo J* 17: 6427–6436, 1998
52. Kolodner RD, Marsischky GT: Eukaryotic DNA mismatch repair. *Curr Opin Genet Dev* 9: 89–96, 1999
53. Marra G, Boland C: Hereditary nonpolyposis colorectal cancer: The syndrome, the genes, and historical perspective. *Journal of the National Cancer Institute* 87: 1114–1125, 1995
54. Wijnen JT, Vasen HF, Khan PM, Zwinderman AH, van der Klift H, Mulder A, Tops C, Moller P, Fodde R: Clinical findings with implications for genetic testing in families with clustering of colorectal cancer. *N Engl J Med* 339: 511–518, 1998
55. Liu B, Parsons R, Papadopoulos N, Nicolaides NC, Lynch HT, Watson P, Jass JR, Dunlop M, Wyllie A, Peltomaki P, de la Chapelle A, Hamilton SR, Vogelstein B, Kinzler KW: Analysis of mismatch repair genes in hereditary non-polyposis colorectal cancer patients [see comments]. *Nat Med* 2: 169–174, 1996
56. Hemminki A, Peltomaki P, Mecklin JP, Jarvinen H, Salovaara R, Nystrom-Lahti M, de la Chapelle A, Aaltonen LA: Loss of the wild type MLH1 gene is a feature of hereditary nonpolyposis colorectal cancer. *Nat Genet* 8: 405–410, 1994
57. Peltomaki P, Vasen HF: Mutations predisposing to hereditary nonpolyposis colorectal cancer: Database and results of a collaborative study. The International Collaborative Group on Hereditary Nonpolyposis Colorectal Cancer, *Gastroenterology* 113: 1146–1158, 1997
58. Yan H, Papadopoulos N, Marra G, Perrera C, Jiricny J, Boland CR, Lynch HT, Chadwick RB, de la Chapelle A, Berg K, Eshleman JR, Yuan W, Markowitz S, Laken SJ, Lengauer C, Kinzler KW, Vogelstein B: Conversion of diploidy to haploidy. *Nature* 403: 723–724, 2000
59. Langer S, Jentsch I, Gangnus R, Yan H, Lengauer C, Speicher MR: Facilitating haplotype analysis by fully automated analysis of all chromosomes in human-mouse hybrid cell lines. *Cytogenet Cell Genet* 93: 11–15, 2001
60. Liu B, Nicolaides N, Markowitz S, Willson J, Parsons R, Jen J, de la Chapelle A, Hamilton S, Kinzler K, Vogelstein B: Mismatch repair gene defects in sporadic colorectal cancers with microsatellite instability. *Nature Genetics* 9: 48–53, 1995
61. Kane M, Loda M, Gaida G, Lipman J, Mishra R, Goldman H, Jessup J, Kolodner R: Methylation of the *hMLH1* promoter correlates with lack of expression of hMLH1 in sporadic colon tumors and mismatch repair-defective human tumor cell lines. *Cancer Research* 57: 808–811, 1997
62. Bellacosa A, Cicchillitti L, Schepis F, Riccio A, Yeung A, Matsumoto Y, Golemis E, Genuardi M, Neri G: MED1, a novel human methyl-CpG-binding endonuclease, interacts with DNA mismatch repair protein MLH1. *Proceedings of the National Academy of Science USA* 96: 3969–3974, 1999
63. Percepe A, Kristo P, Aaltonen L, Ponz de Leon M, de la Chapelle A, Peltomaki P: Mismatch repair genes and mononucleotide tracts as mutation targets in colorectal tumors with different degrees of microsatellite instability. *Oncogene* 17: 157–163, 1998
64. Herman J, Umar A, Polyak K, Graff J, Ahujia N, Issa J-PJ, Markowitz S, Willson J, Hamilton S, Kinzler K, Kane M, Kolodner R, Vogelstein B, Kunkel T, Baylin S: Incidence and functional consequences of *hMLH1* promoter hypermethylation in colorectal carcinoma. *Proceedings of the National Academy of Science USA* 95: 6870–6875, 1998
65. Baylin SB, Herman JG: DNA hypermethylation in tumorigenesis: Epigenetics joins genetics. *Trends Genet* 16: 168–174, 2000
66. Jones P, Laird P: Cancer epigenetics comes of age. *Nature Genetics* 21: 163–167, 1999
67. Deng G, Chen A, Hong J, Chae H, Kim Y: Methylation of CpG in a small region of the hMLH1 promoter invariably correlates with the absence of gene expression. *Cancer Research* 59: 2029–2033, 1999
68. Deng G, Chen A, Pong E, Kim YS: Methylation in hMLH1 promoter interferes with its binding to transcription factor CBF and inhibits gene expression. *Oncogene* 20: 7120–7127, 2001

69. Grady WM, Willis J, Guilford PJ, Dunbier AK, Toro TT, Lynch H, Wiesner G, Ferguson K, Eng C, Park JG, Kim SJ, Markowitz S: Methylation of the CDH1 promoter as the second genetic hit in hereditary diffuse gastric cancer [In Process Citation]. *Nat Genet* 26: 16–17, 2000
70. Esteller M, Toyota M, Sanchez-Cespedes M, Capella G, Peinado MA, Watkins DN, Issa JP, Sidransky D, Baylin SB, Herman JG: Inactivation of the DNA repair gene O6-methylguanine-DNA methyltransferase by promoter hypermethylation is associated with G to A mutations in K-ras in colorectal tumorigenesis. *Cancer Res* 60: 2368–2371, 2000
71. Herman JG, Merlo A, Mao L, Lapidus RG, Issa JP, Davidson NE, Sidransky D, Baylin SB: Inactivation of the CDKN2/p16/MTS1 gene is frequently associated with aberrant DNA methylation in all common human cancers. *Cancer Res* 55: 4525–4530, 1995
72. Toyota M, Ho C, Ahuja N, Jair KW, Li Q, Ohe-Toyota M, Baylin SB, Issa JP: Identification of differentially methylated sequences in colorectal cancer by methylated CpG island amplification. *Cancer Res* 59: 2307–2312, 1999
73. Toyota M, Ahuja N, Ohe-Toyota M, Herman JG, Baylin SB, Issa JP: CpG island methylator phenotype in colorectal cancer. *Proc Natl Acad Sci USA* 96: 8681–8686, 1999
74. Ried T, Knutzen R, Steinbeck R, Blegen H, Schrock E, Heselmeyer K, duManoir S, Auer G: Comparative genomic hybridization reveals a specific pattern of chromosomal gains and losses during the genesis of colorectal tumors. *Genes, Chromosomes and Cancer* 15: 234–245, 1996
75. Phear G, Bhattacharyya N, Meuth M: Loss of heterozygosity and base substitution at the APRT locus in mismatch-repair-proficient and -deficient colorectal carcinoma cell lines. *Molecular and Cellular Biology* 16: 6516–6523, 1996
76. Sato N, Mizumoto K, Nakamura M, Maehara N, Minamishima YA, Nishio S, Nagai E, Tanaka M: Correlation between centrosome abnormalities and chromosomal instability in human pancreatic cancer cells. *Cancer Genet Cytogenet* 126: 13–19, 2001
77. Saunders WS, Shuster M, Huang X, Gharaibeh B, Enyenihi AH, Petersen I, Gollin SM: Chromosomal instability and cytoskeletal defects in oral cancer cells. *Proc Natl Acad Sci USA* 97: 303–308, 2000
78. Amon A: The spindle checkpoint. *Curr Opin Genet Dev* 9: 69–75, 1999
79. Tang Z, Bharadwaj R, Li B, Yu H: Mad2-Independent inhibition of APC^{Cdc20} by the mitotic checkpoint protein BubR1. *Dev Cell* 1: 227–237, 2001
80. Li Y, Benezra R: Identification of a human mitotic checkpoint gene: hsMAD2. *Science* 274: 246–248, 1996
81. Jin D, Spencer F, Jeang K: Human T cell leukemia virus type 1 oncoprotein Tax targets the human mitotic checkpoint protein MAD1. *Cell* 93: 81–91, 1998
82. Dominguez A, Ramos-Morales F, Romero F, Rios RM, Dreyfus F, Tortolero M, Pintor-Toro JA: hpttg, a human homologue of rat pttg, is overexpressed in hematopoietic neoplasms. Evidence for a transcriptional activation function of hPTTG. *Oncogene* 17: 2187–2193, 1998
83. Saez C, Japon MA, Ramos-Morales F, Romero F, Segura DI, Tortolero M, Pintor-Toro JA: hpttg is over-expressed in pituitary adenomas and other primary epithelial neoplasias. *Oncogene* 18: 5473–5476, 1999
84. Heaney AP, Singson R, McCabe CJ, Nelson V, Nakashima M, Melmed S: Expression of pituitary-tumor transforming gene in colorectal tumors. *Lancet* 355: 716–719, 2000
85. Michel LS, Liberal V, Chatterjee A, Kirchwegger R, Pasche B, Gerald W, Dobles M, Sorger PK, Murty VV, Benezra R: MAD2 haplo-insufficiency causes premature anaphase and chromosome instability in mammalian cells. *Nature* 409: 355–359, 2001
86. Cahill DP, da Costa LT, Carson-Walter EB, Kinzler KW, Vogelstein B, Lengauer C: Characterization of MAD2B and other mitotic spindle checkpoint genes. *Genomics* 58: 181–187, 1999
87. Doxsey S: The centrosome—a tiny organelle with big potential. *Nature Genetics* 20: 104–106, 1998
88. Brinkley BR: Managing the centrosome numbers game: From chaos to stability in cancer cell division. *Trends Cell Biol* 11: 18–21, 2001
89. Pihan GA, Purohit A, Wallace J, Malhotra R, Liotta L, Doxsey SJ: Centrosome defects can account for cellular and genetic changes that characterize prostate cancer progression. *Cancer Res* 61: 2212–2219, 2001
90. Lingle WL, Barrett SL, Negron VC, D'Assoro AB, Boeneman K, Liu W, Whitehead CM, Reynolds C, Salisbury JL: Centrosome amplification drives chromosomal instability in breast tumor development. *Proc Natl Acad Sci USA* 99: 1978–1983, 2002
91. Zhou H, Kuang J, Zhong L, Kuo W-I, Gray J, Sahin A, Brinkley B, Sen S: tumor amplified kinase STK15/BTAK induces centrosome amplification, aneuploidy and transformation. *Nature Genetics* 20: 189–193, 1998
92. Sen S, Zhou H, White RA: A putative serine/threonine kinase encoding gene BTAK on chromosome 20q13 is amplified and overexpressed in human breast cancer cell lines. *Oncogene* 14: 2195–2200, 1997
93. Bischoff JR, Anderson L, Zhu Y, Mossie K, Ng L, Souza B, Schryver B, Flanagan P, Clairvoyant F, Ginther C, Chan CS, Novotny M, Slamon DJ, Plowman GD: A homologue of *Drosophila* aurora kinase is oncogenic and amplified in human colorectal cancers. *Embo J* 17: 3052–3065, 1998
94. Glover DM, Leibowitz MH, McLean DA, Parry H: Mutations in aurora prevent centrosome separation leading to the formation of monopolar spindles. *Cell* 81: 95–105, 1995
95. Francisco L, Wang W, Chan CS: Type 1 protein phosphatase acts in opposition to IpL1 protein kinase in regulating yeast chromosome segregation. *Mol Cell Biol* 14: 4731–4740, 1994
96. Myung K, Datta A, Kolodner RD: Suppression of spontaneous chromosomal rearrangements by S phase checkpoint functions in *Saccharomyces cerevisiae*. *Cell* 104: 397–408, 2001
97. Nyberg KA, Michelson RJ, Putnam CW, Weinert TA: Toward maintaining the genome: DNA damage and replication checkpoints. *Annu Rev Genet* 36: 617–656, 2002

98. Bao S, Chang M-S, Auclair D, Sun Y, Wang Y, Wong W-K, Zhang J, Liu Y, Qian X, Sutherland R, Magi-Galluzi C, Weisberg E, Cheng E, Hao L, Sasaki H, Campbell M, Kraeft S-K, Loda M, Lo K-M, Chen L: *HRad17*, a human homologue of the *Schizosaccharomyces pombe* checkpoint gene *rad17*, is overexpressed in colon carcinoma. *Cancer Research* 59: 2023–2028, 1999
99. Rotman G, Shiloh Y: ATM: From gene to function. *Human Molecular Genetics* 7: 1555–1563, 1998
100. Smith L: Duplication of ATR inhibits MyoD, induces aneuploidy and eliminates radiation-induced G1 arrest. *Nature Genetics* 19: 39–46, 1998
101. Zhang H, Tomblin G, Weber B: BRCA1, BRCA2, and DNA damage response: Collision or collusion. *Cell* 92: 433–436, 1998
102. Lane D: Awakening angels. *Nature* 394: 616–617, 1998
103. Rouse J, Jackson SP: Interfaces between the detection, signaling, and repair of DNA damage. *Science* 297: 547–551, 2002
104. Kolodner RD, Putnam CD, Myung K: Maintenance of genome stability in *Saccharomyces cerevisiae*. *Science* 297: 552–557, 2002
105. Ochiai A, Hirohashi S: Multiple genetic alterations in gastric cancer. In: Sugimura T, and Sasako M (eds) *Gastric Cancer*, pp. 87–99. Oxford University Press, New York, 1997
106. Somasundaram K: Tumor suppressor p53: Regulation and function. *Front Biosci* 5: D424–D437, 2000
107. Lane DP: Cancer. A death in the life of p53. *Nature* 362: 786–787, 1993
108. el-Deiry WS, Harper JW, O'Connor PM, Velculescu VE, Canman CE, Jackman J, Pieterpol JA, Burrell M, Hill DE, Wang Y, et al. WAF1/CIP1 is induced in p53-mediated G1 arrest and apoptosis. *Cancer Res* 54: 1169–1174, 1994
109. el-Deiry WS, Tokino T, Velculescu VE, Levy DB, Parsons R, Trent JM, Lin D, Mercer WE, Kinzler KW, Vogelstein B: WAF1, a potential mediator of p53 tumor suppression. *Cell* 75: 817–825, 1993
110. Smith ML, Chen IT, Zhan Q, Bae I, Chen CY, Gilmer TM, Kastan MB, O'Connor PM, Fornace AJ, Jr. Inter-action of the p53-regulated protein Gadd45 with proliferating cell nuclear antigen. *Science* 266: 1376–1380, 1994
111. Lin D, Shields MT, Ullrich SJ, Appella E, Mercer WE: Growth arrest induced by wild-type p53 protein blocks cells prior to or near the restriction point in late G1 phase. *Proc Natl Acad Sci USA* 89: 9210–9214, 1992
112. Bunz F, Fauth C, Speicher MR, Dutriaux A, Sedivy JM, Kinzler KW, Vogelstein B, Lengauer C: Targeted inactivation of p53 in human cells does not result in aneuploidy. *Cancer Res* 62: 1129–1133, 2002
113. Glick AB, Weinberg WC, Wu IH, Quan W, Yuspa SH: Transforming growth factor beta 1 suppresses genomic instability independent of a G1 arrest, p53, and Rb [published erratum appears in *Cancer Res* 1997 May 15;57(10):2079]. *Cancer Res* 56: 3645–3650, 1996
114. Shadan FF, Cunningham C, Boland CR: JC virus: A biomarker for colorectal cancer? *Med Hypotheses* 59: 667–669, 2002
115. Felsher DW, Bishop JM: Transient excess of MYC activity can elicit genomic instability and tumorigenesis. *Proc Natl Acad Sci USA* 96: 3940–3944, 1999
116. Maser RS, DePinho RA: Connecting chromosomes, crisis, and cancer. *Science* 297: 565–569, 2002
117. Kim NW, Piatyszek MA, Prowse KR, Harley CB, West MD, Ho PL, Coviello GM, Wright WE, Weinrich SL, Shay JW: Specific association of human telomerase activity with immortal cells and cancer. *Science* 266: 2011–2015, 1994
118. Rudolph KL, Millard M, Bosenberg MW, DePinho RA: Telomere dysfunction and evolution of intestinal carcinoma in mice and humans. *Nat Genet* 28: 155–159, 2001
119. O'Hagan RC, Chang S, Maser RS, Mohan R, Artandi SE, Chin L, DePinho RA: Telomere dysfunction provokes regional amplification and deletion in cancer genomes. *Cancer Cell* 2: 149–155, 2002
120. Artandi SE, Chang S, Lee SL, Alson S, Gottlieb GJ, Chin L, DePinho RA: Telomere dysfunction promotes non-reciprocal translocations and epithelial cancers in mice. *Nature* 406: 641–645, 2000
121. Hoglund M, Gisselsson D, Hansen GB, Sall T, Mitelman F, Nilbert M: Dissecting karyotypic patterns in colorectal tumors: Two distinct but overlapping pathways in the adenoma-carcinoma transition. *Cancer Res* 62: 5939–5946, 2002